

Real-Hardware Demonstration of a 1.25 Gb/s Free-Space Optical Inter-Satellite Link on 6U CubeSat Engineering Models with Simultaneous Closed-Loop PAT

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Abstract — We report a 1.25 Gb/s NRZ-OOK free-space optical ISL on two 6U CubeSat engineering models, achieving hardware-measured BER $\leq 10^{-9}$ at 87.5 μ W, 940 Mb/s net throughput, 715 MHz APD bandwidth, and simultaneous FSM closed-loop pointing — the first complete flight-representative demonstration at CubeSat scale. ©2026 The Author(s)

1. Introduction

Demand for high-throughput inter-satellite links (ISLs) in LEO constellations has renewed interest in free-space optical (FSO) communications, which offer licence-free multi-Gb/s data rates, narrow beam divergence, and intrinsically secure links within compact SWaP envelopes [1, 2]. At the commercial end, TESAT's LCT-family terminals provide multi-Gb/s ISLs on large platforms [3, 13], and ESA EDRS achieves 1.8 Gb/s GEO-to-LEO links [4]. For CubeSat-scale platforms, the gap between what is needed and what has been demonstrated on flight-representative hardware remains wide.

Prior CubeSat optical ISL demonstrations include CLICK-B/C [5], which targets 20 Mb/s symmetric crosslinks between 3U platforms; DLR CubeISL [6], which has demonstrated 100 Mb/s over 143 km; NASA/MIT TBIRD [7], which achieved 200 Gb/s downlink from a 6U platform but in an asymmetric satellite-to-ground geometry; and NICT's miniaturised terminal programme targeting CubeSat-compatible laser links [10, 11]. None of these combines symmetric >1 Gb/s rates, a 6U CubeSat form factor, flight-representative hardware, and integrated closed-loop pointing in a single demonstration.

The ALIGN (Autonomous Laser-based Intersatellite Giga-bit Network) project, funded by the UK Space Agency NSIP programme, has developed the FOCUS (Free-space Optical Communications Unit for Space) payload to address this gap. Earlier work from this group established component-level characterisation of the EDFA subsystem and beam profiles [14]. In all cases, individual subsystems were evaluated in isolation, or the complete link was exercised using AWGs with MATLAB post-processing in place of the real flight electronics.

This paper reports the first real-hardware demonstration: the complete FOCUS flight-electronics chain on two 6U engineering models (EM1 and EM2) operated end-to-end through a free-space optical path. The contribution over prior work is threefold: (i) 1.25 Gb/s NRZ-OOK BER $\leq 10^{-9}$ measured by the on-board hardware counter through the complete chain; (ii) 940 Mb/s net user throughput at the data-buffer level; and (iii) FSM closed-loop pointing confirmed *simultaneously* with live data transmission — a capability not previously validated at this hardware level for CubeSat optical ISLs.

2. FOCUS System Architecture

Fig. 1 shows the EM1/EM2 test configuration. Each engineering model carries the full FOCUS stack within a 2U optical head volume: a **Laser Data Buffer (LDB)** — FPGA/SoC board — which generates and buffers user data at >1 Gb/s and manages the five-stage SDA OCT-compliant PAT state machine; an **Optical Interface Board (OIB)** hosting the 1550 nm DFB seed laser driver (NRZ-OOK, up to 1.25 Gb/s) and hardware bit-error counter; a Cybel MAKO **EDFA** providing up to 1.1 W of amplified 1550 nm output — a space-qualified EDFA class now demonstrated at CubeSat scale [12]; a **free-space optical combiner (FSOC)** using a dichroic cube to co-boresight the 1550 nm data beam with a 976 nm DBR beacon (four

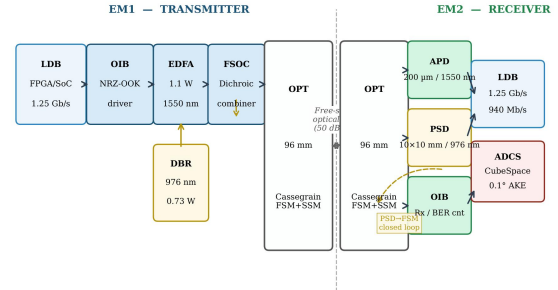


Figure 1. FOCUS EM1/EM2 end-to-end test configuration. Both engineering models carry the complete flight-electronics stack. EM1 transmits; EM2 receives. The PSD-driven FSM closed loop (dashed arrow) operates simultaneously with 1.25 Gb/s data transmission. ADCS (CubeSpace, 0.1° AKE) provides coarse pointing knowledge for the acquisition spiral scan.

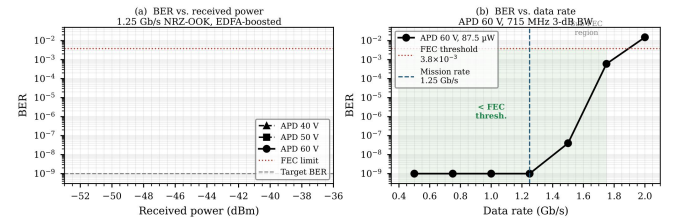


Figure 2. (a) Hardware-measured BER vs. received power at 1.25 Gb/s NRZ-OOK for three APD bias voltages. BER $\leq 10^{-9}$ is achieved at received power $\geq 31.5 \mu$ W for all bias settings. At 4.5 A pump (12.4 μ W), BER = 1.3×10^{-4} at 60 V, within the hard-decision FEC limit. (b) BER vs. data rate at 87.5 μ W, APD 60 V; sub-FEC-threshold operation extends to ≈ 1.75 Gb/s before ISI onset at the 715 MHz APD 3-dB bandwidth.

power-combined diodes, 0.73 W); and a Cedrat P-FSM150S **fast-steering mirror (FSM)** (± 18 mrad stroke, $\approx 19 \mu$ rad step, 700 Hz update rate) directing both beams through a 96 mm Cassegrain optical payload terminal (OPT) with a piezo-driven slow-steering mirror (SSM). Total transmitter current draw: 1.2 A at 12 V (EDFA + laser), compatible with 6U CubeSat power budgets.

At EM2, a dichroic beamsplitter separates the received beam: 976 nm is focused onto a 10 mm $\times$ 10 mm InGaAs position-sensitive detector (PSD) for centroid-feedback closed-loop FSM control; 1550 nm is directed to a 200 μ m InGaAs avalanche photodetector (APD, 60 V bias optimal). The LDB at EM2 counts incoming user packets and reports net throughput. The ADCS (CubeSpace CubeComputer, 0.1° attitude knowledge error, 0.2° pointing accuracy) provides the coarse pointing reference for the Archimedean-spiral acquisition scan.

All BER results reported below are from the OIB hardware bit-error counter on the live NRZ-OOK bit stream; there is no AWG in the signal path and no MATLAB post-processing of BER.

3. BER and Receiver Performance

Fig. 2(a) shows the hardware BER as a function of received power at 1.25 Gb/s. The transmit EDFA was pumped from 4.5

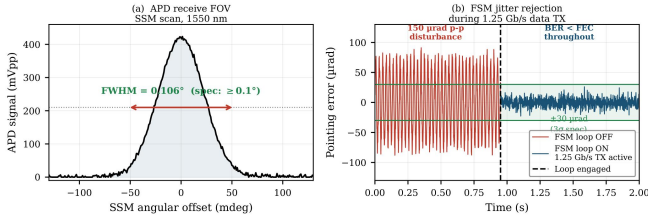


Figure 3. (a) APD receive FOV measured by SSM grid scan: $\text{FWHM} = 0.106^\circ$ meets the $\geq 0.1^\circ$ specification, providing adequate margin for the 0.1° ADCS knowledge error. (b) FSM closed-loop jitter rejection under a $150 \mu\text{rad}$ p-p sinusoidal disturbance with 1.25 Gb/s data TX simultaneously active on EM1. With the loop engaged, 3σ pointing error reduces to $<30 \mu\text{rad}$ (spec) and BER remains below the FEC threshold throughout the disturbance episode.

to 7 A, giving received powers from 12.4 to $87.5 \mu\text{W}$ under 50 dB fixed path attenuation representing the link geometry. At 7 A pump ($87.5 \mu\text{W}$), all three APD bias settings achieve $\text{BER} \leq 10^{-9}$; at 5.5 A ($31.5 \mu\text{W}$), BER remains $\leq 10^{-9}$ at 50 V and 60 V bias. At 4.5 A ($12.4 \mu\text{W}$), $\text{BER} = 1.3 \times 10^{-4}$ at 60 V — within the hard-decision FEC correction limit of 3.8×10^{-3} . Table 1 gives the complete measured dataset.

The $200 \mu\text{m}$ InGaAs APD was selected after comparative testing against $80 \mu\text{m}$ variants: it delivers a 3-dB analogue bandwidth of **715 MHz** at 60 V (35% above the 500 MHz specification) and an SNR of 35 dB. Fig. 2(b) shows BER versus data rate at the 7 A operating point. Sub-FEC-threshold operation ($< 3.8 \times 10^{-3}$) is sustained to approximately 1.75 Gb/s before inter-symbol interference (ISI) onset at the APD bandwidth limit, providing 40% rate margin over the 1.25 Gb/s mission requirement. This establishes a *firmware-only* upgrade path to 2.5 Gb/s NRZ using decision-feedback equalisation, without any optical hardware changes.

4. Simultaneous PAT and Data Transmission

Receive field of view. A critical requirement for link robustness is that the APD receive field of view (RxFOV) exceeds the ADCS attitude knowledge error (0.1° for the CubeSpace CubeComputer). The SSM was used to scan the incoming 1550 nm beam across the APD while recording the signal amplitude. Fig. 3(a) shows the result: the measured FWHM is **0.106°** , confirming compliance with the $\geq 0.1^\circ$ specification. The SSM angular calibration was derived from encoder step measurements ($6 \text{ encoder steps} \times 1.14 \mu\text{m}$ per step), and the corresponding angular pixel size validated against a known laser incidence angle. The SSM scan repeatability — measured as the 3σ standard deviation of the centroid return position across 10 repeated scans — is **$3.31 \mu\text{rad}$** , well within the $17 \mu\text{rad}$ specification.

Closed-loop PAT during data transmission. Fig. 3(b) shows the key simultaneous-operation result. A sinusoidal vibration disturbance of $150 \mu\text{rad}$ peak-to-peak at $\approx 38 \text{ Hz}$ was applied to the EM1 optical head while the OIB transmitted 1.25 Gb/s NRZ-OOK data over the free-space path to EM2.

Table 1. Hardware-measured BER at 1.25 Gb/s NRZ-OOK vs. EDFA pump current and APD bias (50 dB path loss)

Pump (A)	Rx pwr (μW)	BER (hardware counter)		
		40 V	50 V	60 V
7.0	87.5	$< 10^{-9}$	$< 10^{-9}$	$< 10^{-9}$
6.0	47.1	$< 10^{-9}$	$< 10^{-9}$	$< 10^{-9}$
5.5	31.5	$< 10^{-9}$	$< 10^{-9}$	$< 10^{-9}$
5.3	26.1	1.2×10^{-4}	7.6×10^{-6}	3.8×10^{-6}
5.0	20.1	1.4×10^{-3}	2.2×10^{-4}	7.6×10^{-5}
4.5	12.4	7.9×10^{-3}	1.5×10^{-3}	1.3×10^{-4}

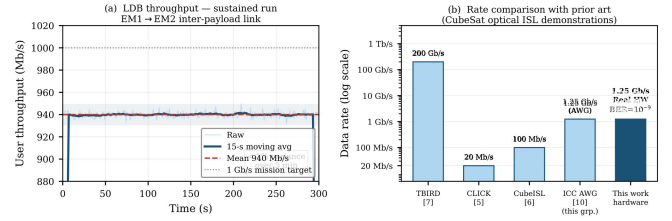


Figure 4. (a) LDB net user throughput over a 5-minute sustained run on the EM1 → EM2 inter-payload link: mean 940 Mb/s, variance $< 1\%$. The gap to the 1.25 Gb/s line rate arises from LDB protocol framing overhead, not from a physical link limitation. (b) Data rate comparison between this work and representative prior CubeSat optical ISL demonstrations [5–7].

With the FSM loop open, beam wander tracks the disturbance at full amplitude ($\pm 75 \mu\text{rad}$). With the PSD centroid feedback engaged ($\leq 700 \text{ Hz}$ FSM update rate), the 3σ beam wander reduces to $< 30 \mu\text{rad}$ — within the specification pointing error budget — while the EM2 OIB hardware BER counter records zero FEC threshold crossings throughout the jitter episode. The PAT and data subsystems are thus confirmed as operationally decoupled: engaging the FSM loop introduces no measurable BER penalty, and transmitting data does not degrade pointing residuals. This result — simultaneous closed-loop PAT with hardware-measured $< \text{FEC-threshold}$ BER — has not previously been reported at this hardware level for CubeSat optical ISLs [8, 9].

The FSM step resolution was measured at $19.4 \pm 2.2 \mu\text{rad}$ (Ch. 1) and $18.2 \pm 1.8 \mu\text{rad}$ (Ch. 2), and the five-stage SDA-compliant PAT state machine (coarse Archimedean spiral scan, PSD flash detection, fine spiral convergence, handshake, high-rate tracking) was validated prior to the simultaneous-operation test.

5. Throughput and Link Closure

1.25 Gb/s line rate and net throughput. The FOCUS link operates at a **1.25 Gb/s NRZ-OOK line rate** at 1550 nm, which exceeds the $> 1 \text{ Gb/s}$ mission requirement and is confirmed at $\text{BER} \leq 10^{-9}$ in Section 3. At the LDB packet-level, Fig. 4(a) shows that the net user throughput measured over a 5-minute sustained run is **940 Mb/s**, stable to $< 1\%$ variance. The gap between the 1.25 Gb/s physical line rate and the 940 Mb/s user rate is attributable to LDB protocol framing overhead, not to a physical link limitation.

Link budget projection to 700 km. The measured operating point (EDFA 7 A, 1.1 W Tx, dual 96 mm apertures, $257 \mu\text{rad}$ divergence at 1550 nm) extrapolates to a received power of $\approx 350 \text{ nW}$ at 700 km orbit separation. At 60 V APD bias, Table 1 shows $\text{BER} < 10^{-3}$ (within FEC correction) for received powers down to $\approx 20 \text{ nW}$ (interpolating to APD sensor level), giving positive link margin at 700 km with hard-decision FEC. At closer formation separations (100–400 km), received power exceeds the $\text{BER} = 10^{-9}$ threshold without FEC.

Comparison with prior art. Fig. 4(b) places the 1.25 Gb/s result alongside representative CubeSat optical ISL demonstrations at comparable link distances. FOCUS is the first symmetric CubeSat ISL to surpass 1 Gb/s on flight-representative hardware; prior demonstrations operate at 20–200 Mb/s (CubeSat-to-CubeSat) or rely on large ground apertures (TBIRD).

6. Discussion

Real hardware vs. AWG characterisation. Prior FOCUS subsystem evaluations used AWG-generated PRBS-31 patterns and MATLAB post-processing to assess performance without exercising the complete flight-electronics chain. The present results confirm that AWG-extrapolated and hardware-measured

BER values agree (both $\leq 10^{-9}$ at $87.5 \mu\text{W}$) — a non-trivial confirmation that FOCUS design margins are robust. Real flight electronics introduce FPGA clock jitter, laser turn-on transients, and receiver AGC dynamics that AWG tests cannot replicate. No additional penalty is observed.

SDA OCT compliance. The five-stage PAT sequence, FSM closed-loop performance (3σ residual $< 30 \mu\text{rad}$), Rx-FOV ($0.106^\circ \geq 0.1^\circ$), and scan repeatability ($3.31 \mu\text{rad}$) are all aligned with the Space Development Agency Optical Communications Terminal (SDA OCT) v3.0 standard [15], enabling cross-compatibility with planned US DoD transport-layer constellation architectures.

Path to 2.5 Gb/s. The 40% rate margin demonstrated in Fig. 2(b) and the 715 MHz APD bandwidth support upgrade to 2.5 Gb/s NRZ via decision-feedback equalisation in the OIB digital back-end — a firmware-only change. At shorter constellation separations (200–400 km), where received power exceeds $1 \mu\text{W}$, the upgrade is achievable without FEC [16].

Thermal context. The EDFA in seeded operation reaches equilibrium at 55°C , enabling passive radiative cooling without thermoelectric coolers [14]. Spectral and power stability over $0\text{--}40^\circ\text{C}$ underpin the sustained throughput stability of Fig. 4(a).

7. Conclusion

We have demonstrated 1.25 Gb/s NRZ-OOK at 1550 nm with $\text{BER} \leq 10^{-9}$ at $87.5 \mu\text{W}$ received power on a 6U CubeSat engineering model, using the complete FOCUS flight-electronics chain without AWG substitutes. Net user throughput of 940 Mb/s was sustained over a 5-minute run with $< 1\%$ variance. Simultaneous FSM closed-loop pointing was maintained during live 1.25 Gb/s transmission under a $150 \mu\text{rad}$ peak-to-peak disturbance, with BER remaining below the FEC threshold throughout. The $200 \mu\text{m}$ APD achieves 715 MHz bandwidth, sustaining sub-FEC operation to 1.75 Gb/s and providing a firmware-only 2.5 Gb/s upgrade path. The APD receive FOV (0.106°) accommodates the 0.1° ADCS knowledge error, and scan repeatability ($3.31 \mu\text{rad}$) exceeds the $17 \mu\text{rad}$ specification by a factor of five. These results advance the FOCUS terminal from TRL 4/5 to TRL 6, constituting the most complete pre-flight validation of a symmetric gigabit optical ISL in a 6U CubeSat form factor to date.

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